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Automated multi-class ECG arrhythmia detection using VMD and multi-task optimization

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Electrocardiogram (ECG) classification is essential for accurately detecting and tracking heart rhythm disorders. This study proposes a multi-class ECG classification framework for identifying cardiac arrhythmias like Atrial Fibrillation (AF), Ventricular Fibrillation (VF), Normal Rhythm (NR), and Ventricular Tachycardia (VT). The ECG signals were decomposed using Variational Mode Decomposition (VMD), and higher-order statistics as well as entropy-based features were extracted from each mode. Multi-task Particle Swarm Optimization (MT-PSO) was employed to reduce redundant features and enhance the discriminative capability of the dataset. Multiple machine-learning models were evaluated, and optimized feature set led to clear performance improvements. The best results were obtained using LightGBM (ACC 0.993), HistGradientBoost (0.991), XGBoost (0.990), and ExtraTrees (0.990). Execution time also decreased for several models after optimization. Confusion-matrix and ROC analyses confirmed reliable detection across all four cardiac classes, and comparison with reported works shows that the proposed framework offers competitive or improved performance for ECG classification.

Keywords ECG, Multi-task particle swarm optimization, Machine-learning, Cardiac arrhythmias

The heart is among the most basic and essential organs. Blood is pumped to numerous networks of vessels by this engine. In order to pump oxygen and nutrients into the body and remove waste products, the heart beats 100,000 times a day. Electrocardiograms (ECGs) record the electrical actions produced by the heart's beating and measure them on the surface of the body. By recording the electrical activity as PQRST waves, skin electrodes reveal how each chamber functions[1]. Therefore, the cardiac behavior is revealed by morphology that are taken from the ECG signal. The ECG signal, which analyzes the behavior of the heart, gives precise information about the heart. Consequently, cardiac arrhythmia is indicated by an irregular ECG that is either faster or slower than usual. Arrhythmias can have a variety of effects, including immediate life-threatening conditions (ventricular fibrillation and tachycardia), long-term risks, or even death, which is why they are the most common cause of death worldwide.

ECG signal consist of several important waves, including P, QRS, along with T, as well as other physiologically pertinent elements. The P-wave denotes atrial depolarization, the QRS complex indicates ventricular depolarization, alongside T-wave depicts ventricular repolarization[1]. Every component in ECG exhibits traits associated with the function of each cardiac chamber. Therefore, alterations in these regions' anatomy could be a sign that the heart has cardiac abnormalities. An in-depth ECG study is the main and fastest way to realize the presence of coronary artery disease[2]. Moreover, the visual ECG examination and identification of CVD by a cardiologist are time-consuming and irritating processes, which may lead to reduced accuracy (Acc)[3].

Heart rhythms that are irregular or abnormal are a sign of arrhythmia. Under normal circumstances, the heart beats in a steady sinus rhythm, driven by electrical signals originating from sinoatrial node in right atrium[4]. When these electrical impulses become erratic or disrupted, arrhythmias can occur-often silently-posing a risk of serious cardiovascular complications. Timely and accurate detection is essential, as early intervention can

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significantly reduce the likelihood of heart-related illnesses[5]. Diagnosing arrhythmias is essential for the timely detection and treatment of cardiac disorders. It helps medical professionals identify subjects who might require additional assessment, care to prevent arrhythmia-related issues like a stroke, heart failure, or unexpected cardiac death. By creating an accurate assessment procedure, this study aims to enhance the quality of life for individuals suffering from cardiac arrhythmias[6]. Automating identification of cardiac arrhythmia (CA) may alleviate the cardiologist's effort in the evaluation process by improving Acc. Despite various methods for distinguishing CA beats, there exists an opportunity for novel artificial intelligence-driven frameworks that enable swift, accurate, and robust automatic categorization.

CA is an irregular heartbeat that causes problems with the cardiovascular system. Cardiovascular disorder causes a problem with electrical conduction all over the heart[7,8]. A variety of arrhythmias may occur due to insufficient electrical conduction within the heart. Atrial fibrillation (AF), ventricular fibrillation (VF), and ventricular tachycardia (VT) are the three most important arrhythmias that can happen in the cardiovascular system[5]. Atrial fibrillation (AF) is a common arrhythmia that affects 2.3% of adults. It makes it very hard for patients to live a normal life and can also cause a number of problems, such as stroke and thrombosis. The World Health Organization (WHO) says that about 2.3 million Americans and more than seven million Europeans have AF. The AF population is expected to double or even grow more in the next few decades[3,9–11]. An elevated resting heart rate triggers a disrupted electromagnetic impulse that results in AF. The rapid and extensive electric impulses reduce the contraction of the atria. An irregular and sudden contraction of the ventricles occurs due to the uncontrolled firing of the AV node. AF is often temporary, yet it occasionally persists even with intervention[12]. It could raise the likelihood of heart failure and ischemic strokes, which may result in higher mortality rates[13,14]. Non-Sustained Beats in VF is a cardiac rhythm disorder marked by asynchronous electrical impulses, leading to inappropriate tremors of the heart's ventricles and impairing the heart's ability to efficiently pump blood throughout the body. The primary reason for death resulting from cardiac arrest is VF, a medical condition that requires immediate attention[15,16]. A cardiac rhythm called VT commences in the ventricles and has a pulse rate over 100 beats every minute and a QRS duration over 120 ms. Severe hypotension can kill. Failing to address VT can lead to fatal cardiac events[17]. Due to changes in ECG anatomy alongside rhythm from individual biological characteristics, automatic categorization of ECG is immense task[18,19]

Medical professionals or computer algorithms analyze ECG waves for specific patterns or abnormalities in order to diagnose arrhythmias. In addition to machine learning models, a variety of other approaches and techniques are used, including observation, qualitative evaluation, and the use of automated algorithmic methods. These methods aid in identifying morphological transformations in ECG waves or erratic RR intervals that may indicate arrhythmia[20,21].

Analysis of ECG signals alongside identification of diseases by cardiologists can be influenced by human error. This prompted examiners to create prototype for detecting CAs utilizing machine learning (ML) techniques[22]. ML approaches have been successfully used for the identification of cardiac arrhythmias for the past ten years[23]. ML techniques have demonstrated satisfactory results in feature selection (FS) and noise reduction, leading to enhanced ACC in identifying instances of CA. As a result, numerous researchers have put forth various techniques for the automatic identification of CA conditions.

Several methods have been documented in ECG signal analysis to effectively identify CAs[24–27]. These methods detected irregular heart rhythms like VT and VF by using temporal and spectral features. Wavelet analysis was used in a study report to categorize 63 surface ECG segments from MIT-BIH dataset into three groups: VT, OVF and DVF (organized and disorganized VF))[24]. Using two-stage binary classifier, they achieved an Acc of 93.7% for VT identification and 80.0% for OVF alongside DVF separation, respectively. EMD-based technique for identification of VF conditions in a 3-second ECG slices was proposed; it achieved a 99.70% Acc rate for the use of external defibrillators[28]. An innovative algorithm described using discrete wavelet transformation to automatically distinguish between shockable and non-shockable ventricular arrhythmias from ECG signals with lengths of 5,8, and 10secs. In a 10-second ECG segment, the system got best outcomes, with 98.32% Acc[29].

According to[23], ML techniques have been successfully used for the identification of CAs for the past ten years. Decision trees[30,31], Bayesian decisions[28], extreme learning machines[32], and linear discriminant analysis[33,34]. However, support vector machines (SVM)[35–37], and artificial neural networks[38,39] are the most widely used techniques. These methods classify arrhythmia conditions by using 2-D based attributes that are extracted from filtered ECG and fed to classifier. It demonstrated how the multi-feature-based methods significantly improve the VT/VF recognition Acc[40]. It has been proposed that using ML techniques in conjunction with temporal/morphological, spectral, or complexity parameters can effectively increase the detection approach's efficiency[41,4223]. presents a new method for categorizing VF Vs. non-VF rhythms techniques via 13 ECG parameters that integrate temporal, spectral, alongside complexity attributes. The SVM classifier was used to select and categorize the useful features. The method produced a 95% sensitivity (S_e) and 99% specificity (S_p) for shockable and a 95% S_e and 99% S_p for VF, respectively. SVM-based machine learning techniques were used to classify VT/VF via 14 best attributes that were chosen by genetic procedure over range of window lengths from 1 to 10 s[4344]. obtained 96.79% accuracy by directly converting ECG signals into PNG format and methodically sequentially incorporating each lead data using Vision Transformer (ViT) models[45]. suggested a hybrid parallel-residual architecture that integrates residual and a parallel arrangement with bidirectional long short-term memory (Bi-LSTM) and is especially designed for arrhythmia detection. With an accuracy of up to 98.75%, this architecture guarantees reliable feature extraction and accurate classification. A polymorphic representation-aware network (PRA-Net) is introduced in[46]for the detection of cardiac arrhythmias from ECG signals. The adaptive wavelet-aware (AWA) module is designed to recognize ECG signals across various scales by learning multiscale features, achieving an AUC of 98.86%[47]. introduced a QRS-centric methodology that combines convolutional layers for feature extraction alongside bidirectional LSTM layers focused on the QRS complex region via adaptive beat segmentation, achieving a precision of 98.606%. By integrating Residual

Networks (ResNet) for effective feature extraction alongside Bidirectional Long Short-Term Memory (Bi-LSTM) for capturing temporal dependencies, an F1 score of 98.01% was attained. A self-attention autoencoder based on modified Kalman filters improved ECG denoising and achieved 99.71% accuracy[4849]. employed a hybrid CNN BLSTM framework with Mish activation to detect CA automatically with minimum preprocessing. Adaptive neural network classifiers (ANNC), SVM, alongside bagging classifiers (BAGG) were used to classify VF alongside VT conditions via 2D depiction of ECG. According to the results, the specificity and sensitivity for VF detection were 98.8% and 95.56%, for VT detection they were 99.5% and 88.80%, and for normal sinus rhythms they were 97.7% and 98.98%[50].

By employing features derived from decomposed ECG signals, empirical mode decomposition (EMD) produced superior results in discriminating VF conditions[51]. A random forest (RF) classifier was used to classify the signals after they had first been fragmented down by variational mode decomposition (VMD) alongside Renyi entropy, energy, and various parameters were retrieved[26]. reported that the system's Acc, specificity, alonside sensitivity were 97.23%, 97.97%, and 96.54%, respectively[52]. used a C4.5 classifier alongside 13 time-frequency based attributes to identify VT and VF conditions, and they were successful in achieving a 97.03% Acc rate.

Estimating the effectiveness of a attribute set from ECG derived characteristics despite first training and testing the classification algorithm is inherently challenging. In order to achieve the best classification performance, it is necessary to evaluate several feature sets, making the process of choosing ECG features iterative [8, 9]. Minimizing the number of features while improving the model's performance efficiency is known as ECG FS [25]. In the last two years, a number of studies have investigated FS in ECG processing using metaheuristic algorithms.

It is intrinsically challenging to evaluate the efficacy of attribute set obtained by ECG attribute extraction without first training alongside testing a classification algorithm. In order to achieve the best classification performance, it is necessary to evaluate several feature sets, making the process of choosing ECG features iterative[53,54]. ECG FS refers to the process of reducing the number of features to improve the model's performance efficiency[55]. In the last two years, a number of studies have investigated FS in ECG processing using metaheuristic algorithms. Elastic net (EN), least absolute shrinkage selection operator, GA, and particle swarm optimization (PSO) were all tested in[56]. Popular ML techniques like SVM and random forest (RF) classifiers use the features from the optimization phase. The RF classifier in conjunction with the GA and EN approaches consequently produced improved recognition rates of 95.30% and 94.90%, respectively. For the classification of ECG arrhythmias[57], suggested marine predator algorithm (MPA), which combines machine learning classifiers with metaheuristic optimization. With mean ECG arrhythmia classification Acc of 99.92% alongside sensitivity of 99.81%, the results showed that the integration of the MPA considerably enhanced the performance of all tested classifiers. The Butterfly Optimization Algorithm, MRFO, as well as Emperor Penguin Optimization algorithms are used to process the ECG signals in order to identify arrhythmias[58]. The Butterfly Optimization Algorithm performed best among them, achieving 99.14% Acc. For the classification of ECG arrhythmias[59], examined three FS algorithms: matched selection, Chi-square, and minimum redundancy maximum relevance. The experimental findings show that the suggested feature selection technique, known as "Matched Selection", when combined with SVM classifier attains better than alternative strategies, attaining an Acc of 81.27%.

Overall, the features are greatly optimized by ECG feature extraction and FS. These benefits are especially helpful when identifying ECG arrhythmias using machine learning. Building on this framework, a feature-optimized end-to-end methodology for classifying ECG arrhythmias is put forth.

In this manuscript, we describe a method for categorizing VT, VF, normal (NR), alongside AF. The ECG signal are first preprocessed via Circulant singular spectrum alongside 4 stage Savitzky Golay filter. Filtered signal is decomposed using VMD into Intrinsic Mode Function (IMF), from IMFs Higher order statistics (HOS) and entropy based features were extracted. Multi-task Particle Swarm optimization (MTPSO) FS algorithm is applied on the features extracted to obtain prominent features. Form the prominent attributes extracted are given to the machine learning (ML) classifiers for the classification. The efficacy of the proposed method is assessed in terms of Acc, P_r, R_e, and $F1$.

The main contributions and novelties of this work are summarized as follows:

- A novel SVMD MTPSO hybrid framework is proposed for multi-class ECG arrhythmia classification, integrating signal decomposition and multi-task optimization.
- A MultiTaskPSO-based feature selection strategy is introduced to simultaneously optimize feature subsets across multiple classifiers through knowledge sharing.
- A hybrid fitness function balancing classification accuracy and feature reduction is designed to improve model generalization and computational efficiency.
- Extensive comparative analysis is conducted using multiple ensemble and boosting classifiers, demonstrating significant improvement in execution time while maintaining high classification accuracy.
- The proposed method achieves competitive performance (up to 99.3% accuracy) on publicly available PhysioNet datasets, outperforming several existing approaches.

The remainder of the manuscript is organized as follows: Section "Proposed methodology" consists of the Proposed methodology, Section "Results and discussion" is about Results and discussion and conclusion in the Section "Conclusion".

Proposed methodology

The proposed ECG signal processing and classification framework's general workflow is depicted in the Fig. 1. Raw ECG acquisition is the first step in the methodology's methodical, multi-stage pipeline, which ends with performance evaluation.

To improve signal quality, the raw ECG signal is first put through a preprocessing step where noise and artifacts are eliminated. This stage guarantees that the ECG data used in the analysis that follows is clean and trustworthy.

Variational Mode Decomposition (VMD) is then employed to break down preprocessed signal. The ECG signal is broken down by VMD into a collection of intrinsic mode functions (IMFs), each of which represents oscillatory elements at various frequency ranges. These IMFs effectively represent cardiac dynamics by capturing both high-frequency and low-frequency aspects of the ECG signal.

Entropy-based features and Higher-Order Statistics (HOS) are extracted from the decomposed IMFs. While entropy measures quantify the signal's complexity and irregularity, resulting in a discriminative feature set, HOS features capture non-linear and non-Gaussian characteristics of the signal.

Multi-task Particle Swarm Optimization (PSO) is used for FS in order to further improve classification performance and decrease redundancy. This optimization step reduces computational complexity and improves generalization across various ECG classes by identifying task-specific but shared discriminative feature subsets.

The machine learning classification stage, where appropriate ML classifiers are used for training and testing, receives the chosen optimal features. To provide thorough assessment of suggested framework, system performance is finally assessed using common performance metrics.

Datasets used

In this research, we used datasets from MIT-BIH Malignant Ventricular Ectopy database (MVED), the Creighton University Ventricular Tachyarrhythmia database (CVTD), MIT-BIH arrhythmia database (MITBD), and the physioNet warehouse Atrial Fibrillation Database (MBAFD). Ambulatory ECG recorders were used to collect MBAFD at Boston's BIH. It includes 25 lengthy ECG recordings from a patient with AF. VF instances are present in all 35 ECG data sets in CVTD. NR instances are present in MITBD[60]. Since these databases were recorded at different sampling frequencies, all ECG signals were resampled to a uniform rate of 360 Hz to ensure consistency in analysis and feature extraction. All the databases used in this study are publicly available.

ECG decomposition using VMD

The Empirical Mode Decomposition (EMD) technique is widely employed for recursively breaking down signal into different modes, but it has a few drawbacks, including noise sensitivity and sampling. A non-recursive VMD framework that has the capability of concurrent mode extractions was thus created in order to get around these restrictions. Intrinsic Mode Functions (IMFs) are the amplitude-modulated-frequency-modulated (AM-FM) signals[61], and they are as follows:

$$\mathrm{I}_{MF\mathrm{k}}(\mathrm{t}) = \sum_{\mathrm{k}=1}^{\mathrm{M}} \mathrm{P}_{\mathrm{k}}(\mathrm{t})\,\mathrm{Cos}\left(\sigma_{\mathrm{k}}(\mathrm{t})\right) \tag{1}$$

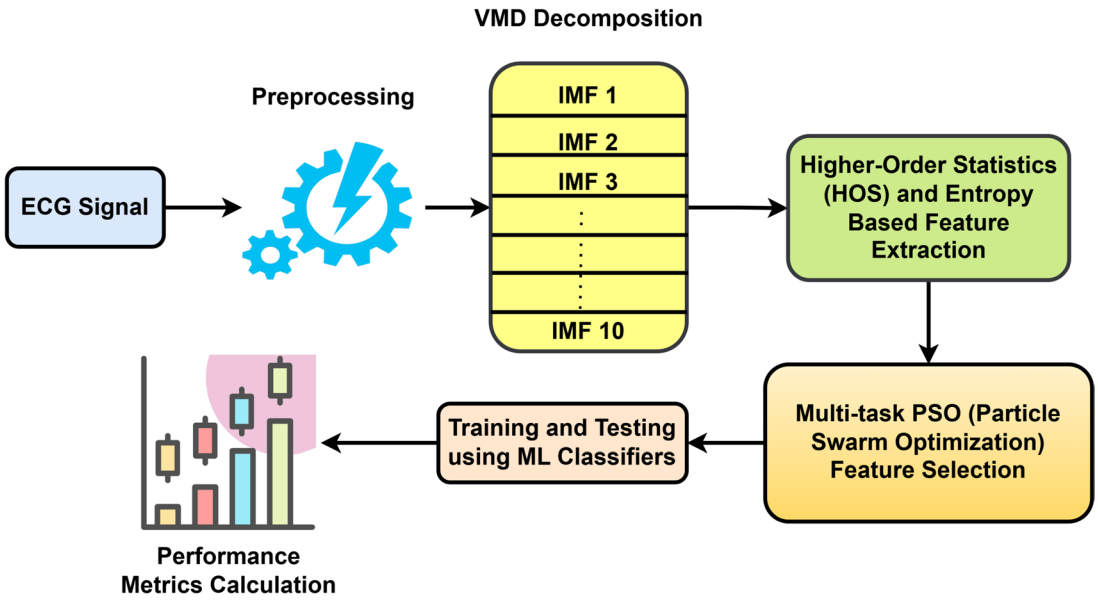


Fig. 1. Overview of the proposed ECG signal processing and classification framework.

where M is number of modes, σ is phase and non-decreasing i.e. $\sigma_k(t) > 0$, and $P_k(t)$ is envelope and non-negative function.

VMD provides a variational alternative to the empirical mode decomposition. VMD's objective is to break down signal into specific number of sub-signals, or IMFs, that can replicate input using their sparsity characteristics[62]. Three parameters are essentially required for the VMD decomposition technique:

- Lagrangian multiplier (η) is used check quality of reconstruction from its IMFs.
- Number of modes (M) has to be configured for to estimate number of decomposed modes.
- Data fidelity constraint parameter (β) to ascertain frequency ranges in various modes.

VMD decomposition algorithm is shown below:

STEP-1: η_1, is started for first iteration, and k^{th} mode's center frequency (w_k) and spectrum (I_{MFk}) are produced.

STEP-2: For all modes i.e. for k = 1 to M,

(i) Spectrum of kth mode in next iteration is updated as:

$$\mathrm{I}_{MF_{\mathrm{k}}}^{\mathrm{n+1}}(\mathrm{t}) = \frac{\mathrm{I}(\mathrm{w}) - \sum_{\mathrm{i}<k} \mathrm{I}_{\mathrm{i}}^{\mathrm{n+1}}(\mathrm{w}) - \sum_{\mathrm{i}>k} \mathrm{I}_{\mathrm{i}}^{\mathrm{n}}(\mathrm{w}) + ((\eta^{\mathrm{n}}(\mathrm{w})/2))}{1 + 2\beta\,(\mathrm{w} - \mathrm{w}_{\mathrm{k}}^{\mathrm{n}})^2} \tag{2}$$

(ii) Center frequency for each mode is updated as:

$$\mathrm{w}_{\mathrm{k}}^{\mathrm{n+1}} = \frac{\int_0^\infty \mathrm{w} \left|\mathrm{I}_{MF_{\mathrm{k}}}^{\mathrm{n+1}}(\mathrm{w})\right|^2 \mathrm{dw}}{\int_0^\infty \left|\mathrm{I}_{MF_{\mathrm{k}}}^{\mathrm{n+1}}(\mathrm{w})\right|^2 \mathrm{dw}} \tag{3}$$

(iii) Update η as follows:

$$\eta^{\mathrm{n+1}} = \eta^{\mathrm{n}} + \tau \left(\mathrm{I}(\mathrm{w}) - \sum_{\mathrm{k}} \mathrm{I}_{MF_{\mathrm{k}}}^{\mathrm{n+1}}(\mathrm{w}) \right) \tag{4}$$

STEP-3: Increase the number of iterations and return to step 2 until convergence is achieved. Following is criteria for convergence:

$$\sum_{\mathrm{k}} \frac{\mathrm{I}_{MF_{\mathrm{k}}}^{\mathrm{m+1}} - \mathrm{I}_{MF_{\mathrm{k2}}}^{\mathrm{m2}}}{\mathrm{I}_{MF_{\mathrm{k2}}}^{\mathrm{m2}}} < \epsilon \tag{5}$$

The decomposition of ECG signal using VMD is shown in the Fig. 2.

Selection of VMD parameters

The performance of VMD is influenced by key parameters, particularly the number of modes (K), the quadratic penalty factor (α), and the convergence tolerance. In this study, these parameters were selected based on a combination of empirical evaluation and stability analysis.

The number of modes K was varied between 1 and 15 to examine its impact on reconstruction quality and class separability. It was observed that lower values of K resulted in under-decomposition, while higher values introduced redundant modes and increased computational complexity. Based on classification performance and spectral separation quality, $K = 10$ was selected as the optimal trade-off.

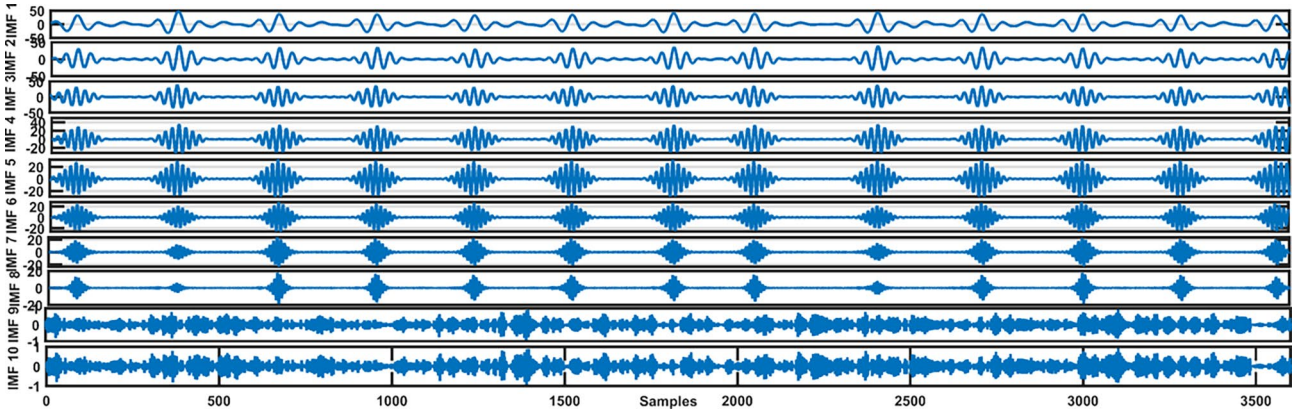


Fig. 2. ECG signal decomposition using VMD.

The penalty parameter α was tuned within the range of 1000 5000. Moderate values ensured adequate bandwidth constraint without oversmoothing intrinsic oscillatory components. The optimal value was selected by monitoring classification accuracy and mode stability across cross-validation folds. The convergence tolerance was fixed at 10^{-7} to ensure stable decomposition without excessive computational overhead.

This parameter selection strategy ensures balanced spectral decomposition, improved feature discrimination, and computational efficiency.

Feature extraction

Feature extraction plays a crucial role in capturing the discriminative characteristics of ECG signals for effective classification. A broad set of features was derived from the intrinsic mode functions (IMFs) obtained through VMD to effectively characterize ECG signals. These features capture information across multiple domains, including irregularity, energy variations, statistical patterns, nonlinear dynamics, and frequency distribution. The extracted features are grouped into five categories: entropy-based, energy-based, statistical, complexity, and spectral features.

Entropy-based features

Shannon Entropy (Sh_{En}): It quantifies average amount of information, uncertainty, or randomness contained in a signal. Mathematically, it is expressed as:

$$Sh_{En} = -\sum_{k=1}^{m} q(x_k) log q(x_k) \tag{6}$$

Approximate Entropy (Ap_{En}): The uncertainty or unpredictability of a finite-length signal is described by Ap_{En}. It is necessary to embed the signal in phase space and calculate the rate at which the number of phase space variations within a given threshold (r) increases as the phase space embedding dimension (m) advances from m to m + 1. To determine the value of Ap_{En}, use the formula below[63]:

$$\mathrm{Ap}_{En}(m, r) = -\ln\left(\frac{1}{N-m+1}\sum_{k=1}^{N-m+1} C_k(m, r)\right) \tag{7}$$

Differential Entropy (Di_{En}): An extension of Shannon entropy for continuous-valued signals, suitable for capturing variability in ECG signals influenced by continuous noise distributions.

$$Di_{En} = -\int_{-\infty}^{\infty} p(x)\log(p(x))dx \tag{8}$$

Rényi Entropy (Re_{En}): The diversity or uncertainty of a probability distribution can be quantified mathematically using the Renyi entropy. The following formula can be used to determine Renyi entropy (Re_{En}):

$$Re_{En} = \frac{1}{1-\alpha}\log_2\sum_{k=1}^{n} p_k^{\alpha} \tag{9}$$

Higher order statistics (HOS) features

Finding trustworthy features that best represent ECG beats is a major challenge in ECG heartbeat classification. Features for effective classification have been successfully extracted using higher order statistics (HOS)[64–69]. The QRS complex's hidden features can be defined by utilizing the HOS of sub-band signals.

The significance of initial and subsequent order statistics in signal processing has grown. However, second-order statistical methods are unable to adequately analyze many signals, particularly with regard to comes to nonlinearities.

HOS techniques have been developed as a result. Higher order statistics include higher order moments (M3, M4,...) along with nonlinear mixtures of higher order moments called cumulants (C1, C2, C3,...), whereas 1st and 2nd order statistics include mean and variance[70].

Let V(n) be a real random process with discrete time. The coefficients in the moment generating function's Taylor series expansion are known as the moments of V(n)[69].

$$\varphi_v(w) = E[\exp(jwv)] \tag{10}$$

The following are the definitions of moments and cumulants for a zero mean discrete time signal:

$$\begin{aligned} M_2(i) &= E[V(n), V(n+i)] \\ M_3(i,j) &= E[V(n), V(n+i)\cdot V(n+j)] \\ M_4(i,j,k) &= E[V(n), X(n+i)\cdot V(n+j)\cdot V(n+k)] \end{aligned} \tag{11}$$

Here E [.] is expectation operation and V(.) is random process.

The second characteristic function of V(n), is given by the equation:

$$V(w) = \ln \varphi_V(w) = \ln E[\exp(jwv)] \tag{12}$$

is termed as cumulant generating function, alnogside coefficients in its Taylor expansion are nth-order cumulants of V(n), depicted as $c_n(\tau_1, \tau_2, ..., \tau_n)$ and cumulants are given by the equations:

$$\begin{aligned} &C_2(i) = M_2(i) \\ &C_3(i,j) = M_3(i,j) \\ &C_4(i,j,k) = M_4(i,j,k) - M_2(i)M_2(j-k) \\ &- M_2(j)M_2(k-i) - M_2(k)M_2(i-j) \end{aligned} \tag{13}$$

For every beat, the $2^{nd}, 3^{rd}, and 4^{th}$ order cumulants are computed using lag 0. The cumulants with zero lag have unique names: $C_2(0) \rightarrow$ variance and is represented by σ^2, $C_3(0,0)$ as well as $C_4(0,0,0)$ are represented by γ_{3v}(skewness), γ_{4v}(kurtosis).

Feature selection
The most crucial step in creating a pattern classification system is identifying the most useful features because even the finest classifier would perform badly if the features are improperly selected.

MultiTaskPSO for feature selection
In MultiTaskPSO, each particle encodes a binary feature mask. Let the position of particle *i* be denoted as:

$$X_i = [x_{i1}, x_{i2}, \ldots, x_{iD}], \qquad x_{id} \in \{0, 1\}, \tag{14}$$

where *D* denotes the number of features. In the proposed MultiTaskPSO framework, each classification model is treated as an independent task. Information sharing between tasks is enabled through global best particle exchange, allowing knowledge transfer while preserving task-specific optimization. The velocity and position of each particle are updated according to standard PSO update rules, and the resulting continuous component z_{id} is obtained prior to binarization.

The updated continuous component z_{id} is mapped to a probability value using a sigmoid transfer function defined as:

$$\sigma(z_{id}) = \frac{1}{1 + e^{-z_{id}}}. \tag{15}$$

The binary feature mask is then obtained as:

$$x_{id} = \begin{cases} 1, & \sigma(z_{id}) > 0.5, \\ 0, & \text{otherwise}, \end{cases} \tag{16}$$

The fitness of each particle is evaluated using a hybrid objective function defined as:

$$F_i = \alpha(1 - Acc_i) + \beta \frac{|S_i|}{D}, \tag{17}$$

where Acc_i denotes the classification accuracy of particle *i*, $|S_i|$ represents the number of selected features in particle *i*, *D* is the total number of features, and α, β are weighting coefficients that balance classification performance and feature reduction. The optimization process terminates when the maximum number of iterations is reached or when no significant improvement in global best fitness is observed.

Multi-task assignment
Each particle is evaluated on all tasks $k \in \{1, \ldots, K\}$ and assigned a skill factor:

$$SF(i) = \arg \min_{k \in \{1,\ldots,K\}} F_i^{(k)}, \tag{18}$$

where $F_i^{(k)}$ denotes the fitness of particle *i* for task *k*.

Fitness function
For FS, the fitness of particle *i* on task *k* is defined as:

$$F_i^{(k)} = 1 - \mathrm{Acc}_k(X_i), \tag{19}$$

where classification accuracy is computed as:

$$\mathrm{Acc}_k(X_i) = \frac{1}{n_k} \sum_{j=1}^{n_k} 1\left(C_k(X_j^{(i)}) = y_j\right), \tag{20}$$

with $1(\cdot)$ being the indicator function.

PSO velocity and position update
For the assigned task $\tau = SF(i)$, the velocity update rule is:

$$V_i(t+1) = \omega V_i(t) + c_1 r_1 \left(pBest_i - X_i(t)\right) + c_2 r_2 \left(gBest_\tau - X_i(t)\right), \tag{21}$$

and the continuous position update follows:

$$Z_i(t+1) = Z_i(t) + V_i(t+1). \tag{22}$$

Binary conversion is applied as:

$$X_i(t+1) = \sigma(Z_i(t+1)). \tag{23}$$

Cross-task knowledge transfer
If random mating probability (RMP) is satisfied, knowledge is exchanged between particles of different tasks:

$$Z_i(t+1) = \frac{Z_i(t+1) + Z_j(t)}{2}, \qquad \text{if rand}() < \text{RMP}. \tag{24}$$

Learning component
For task τ, the refinement is applied as:

$$X_i(t+1) = X_i(t+1) + r_3 \left(X_{\text{best}}^{(\tau)} - |X_i(t+1)|\right) - r_4 \left(X_{\text{worst}}^{(\tau)} - |X_i(t+1)|\right), \tag{25}$$

where $X_{\text{best}}^{(\tau)}$ and $X_{\text{worst}}^{(\tau)}$ denote the best and worst solutions of task τ.

Fitness re-evaluation

$$F_i^{(\tau)}(t+1) = 1 - \text{Acc}_\tau(X_i(t+1)). \tag{26}$$

If improvement is observed:

$$pBest_i = X_i(t+1). \tag{27}$$

Global best update
For each task *k*, the global best is updated as:

$$gBest_k = \arg\min_i F_i^{(k)}. \tag{28}$$

Output
The algorithm returns the optimized feature subsets for all tasks:

$$\{gBest_1, gBest_2, \ldots, gBest_K\}. \tag{29}$$

Convergence behavior of jMultiTaskPSO
In our Multi-Task PSO (JTPSO), a "task" refers to cooperative sub-swarms operating within the same feature selection problem rather than different datasets or objectives. The population is divided into sub-swarms that optimize the same fitness function while maintaining their own local best solutions and sharing information through global and sub-swarm best guidance, along with a feature-weight reinforcement mechanism. We have now included a direct comparison with standard single-task PSO under identical settings, and the results show faster convergence and lower final fitness for JTPSO as in Fig. 3, confirming that the multi-task formulation provides measurable performance gains rather than merely increasing complexity.

Each particle encodes a binary feature mask X_i as defined in (1), derived from a continuous position vector Z_i using the sigmoid-based transfer function in (2). The fitness of every particle for each task is evaluated using (5)–(6), and particles are assigned to tasks via the skill factor mechanism in (4).

During optimization, the velocity update in (7) and the position update in (8) drive particles toward both their personal best ($pBest_i$) and the task-level global best ($gBest_\tau$). After each update, binary conversion (9) and optional cross-task knowledge exchange (10) enable exploration across tasks, while the learning refinement rule in (11) further improves convergence by guiding particles toward the best-performing solutions and away from poor regions of the search space.

As shown in Fig. 3, the best fitness value decreases monotonically with iterations, demonstrating stable and progressive improvement. The curve exhibits multiple plateaus where particles temporarily exploit promising regions, followed by sharp drops caused by successful cross-task knowledge transfer or refinement-based updates. The final fitness stabilizes near 0.299, indicating that the algorithm has converged to an optimal or near-optimal subset of discriminative features. The updates described in (12)–(14) ensure that the best-performing solutions for each task are retained, ultimately yielding the optimized feature subsets $\{gBest_1, gBest_2, \ldots, gBest_K\}$ as expressed in (15). After applying the optimization algorithm, 33 features were selected out of the original

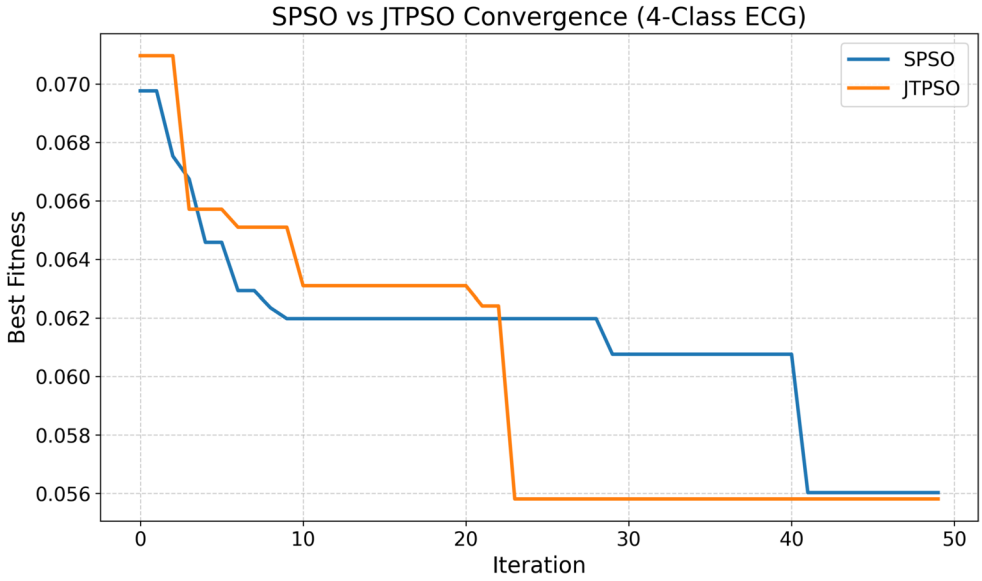


Fig. 3. Fitness curve for MultiTaskPSO optimization.

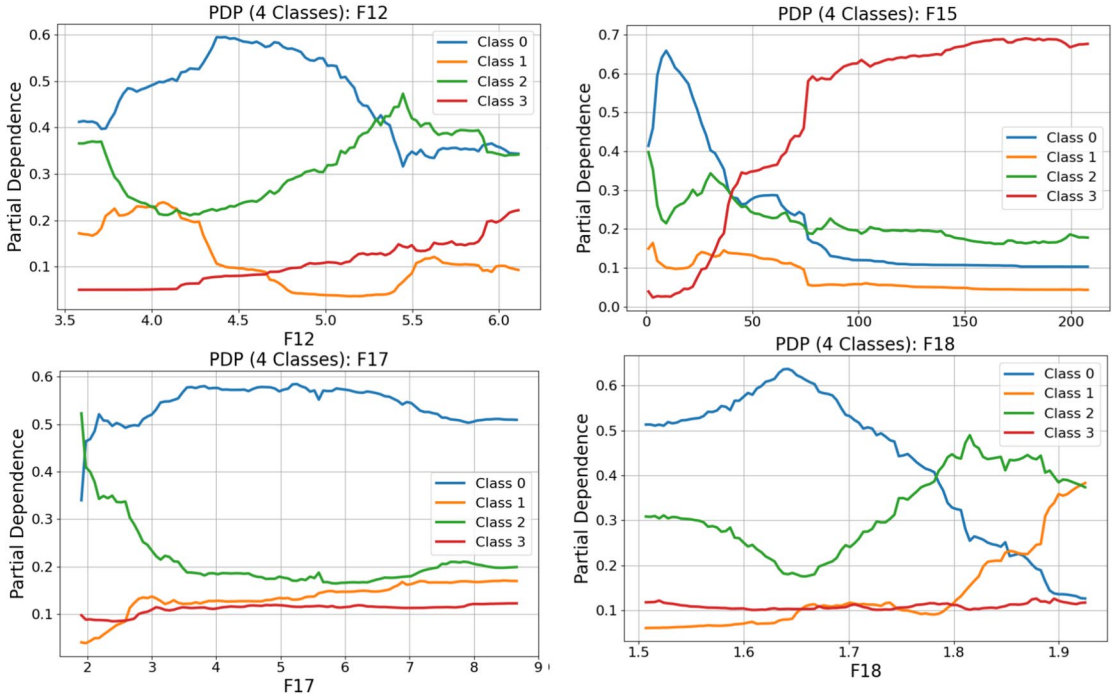


Fig. 4. Interdependency curves for four selected features.

100. These features demonstrate strong inter-dependency and relevance with the target labels, as seen in their dependency curves, which show clear patterns and separability, preserving discriminative power while reducing dimensionality. The Partial Dependence Plot (PDP) analysis for the four most important features (F12, F15, F17, and F18) as in Fig. 4 confirms this: F12 strongly separates Class 0 from the others, F15 shows highly discriminative behavior with Class 3 dominating at higher values and Class 0 decreasing steadily, F17 consistently favors Class 0 across its range, and F18 exhibits nonlinear patterns that help distinguish particularly between Classes 0 and 2. Overall, the smooth, well-separated curves indicate that the model has captured stable and interpretable relationships, confirming good learning behavior and strong discriminative power for the four-class classification problem.

Require: N (particles), T (max iterations), dataset (X, y), threshold τ, PSO params $w, c_1, c_2, c_3, V_{\max}$
Ensure: Selected feature set $S^\star$, best fitness $f^\star$
Initialize $X_{i,j}^{(0)} \sim \mathcal{U}(0,1)$ and velocities $V_{i,j}^{(0)} \leftarrow 0$ for $i = 1 \dots N,\ j = 1 \dots d$
Initialize weight vector $\mathbf{w} \leftarrow \mathbf{0} \in \mathbb{R}^d$
Partition particles into m sub-swarms (e.g., $m = 2$)
for $i \leftarrow 1$ to N **do**
    $m_i \leftarrow \mathbf{1}\{X_{i,:} > \tau\}$ ▷ binary mask
    $f_i \leftarrow \mathcal{F}(m_i)$ ▷ evaluate using wrapper (KNN CV)
    $pbest_i \leftarrow X_{i,:}, \quad pbest_f_i \leftarrow f_i$
    Update sub-swarm bests and global best g accordingly
end for
for $t \leftarrow 1$ to T **do**
    for each particle $i = 1 \dots N$ **do**
        if use PSO-mode **then**
            for each dimension $j = 1 \dots d$ **do**
                Draw $r_1, r_2, r_3 \sim \mathcal{U}(0,1)$
                $v_{i,j} \leftarrow w v_{i,j} + c_1 r_1 (pbest_{i,j} - x_{i,j})$
                $\quad + c_2 r_2 (g_j - x_{i,j}) + c_3 r_3 (s_{k,j} - x_{i,j})$
                Clip $v_{i,j} \in [-V_{\max}, V_{\max}]$
                $x_{i,j} \leftarrow \text{clip}(x_{i,j} + v_{i,j}, 0, 1)$
            end for
        else ▷ binary weight-driven mode
            $\tilde{w} \leftarrow \max(\mathbf{w}, 0)$
            if $\sum_j \tilde{w}_j > 0$ **then**
                for each dimension j **do**
                    set $x_{i,j} \leftarrow 1$ with probability $\tilde{w}_j / \sum \tilde{w}$, else 0
                end for
            else
                perform PSO-mode (fallback)
            end if
        end if
        $m_i^{\text{new}} \leftarrow \mathbf{1}\{X_{i,:} > \tau\}$
        $f_i^{\text{new}} \leftarrow \mathcal{F}(m_i^{\text{new}})$
        if $f_i^{\text{new}} < pbest_f_i$ **then**
            $\Delta \leftarrow pbest_f_i - f_i^{\text{new}}$
            Update per-feature weights:
              increase $w_j \leftarrow w_j + \Delta$ for features j with $m_j^{\text{old}} = 0 \wedge m_j^{\text{new}} = 1$
              decrease $w_j \leftarrow w_j - \Delta$ for features j with $m_j^{\text{old}} = 1 \wedge m_j^{\text{new}} = 0$
            $pbest_i \leftarrow X_{i,:},\ pbest_f_i \leftarrow f_i^{\text{new}}$
        else
            $\Delta \leftarrow f_i^{\text{new}} - pbest_f_i$
            Apply opposite sign adjustments to $\mathbf{w}$ (penalize emerged features)
        end if
        Update sub-swarm best and global best if needed
    end for
    Record $curve[t] \leftarrow f^\star$ and number of features $|\{j : g_j > \tau\}|$
end for
return $S^\star = \{j : g_j > \tau\},\ f^\star$

Algorithm 1. jMultiTaskPSO for Feature Selection

Classifiers

We have experimented with ML classification both with and without FS algorithms to validate the effectiveness of our proposed method.

ML classification with classifiers usually means improving the prediction performance and the generalization power of the system by combining several base learners. This approach exploits capabilities of individual models while at the same time alleviating their drawbacks by a combination strategy like bagging, boosting, or stacking.

ML classification with FS is very important in the process of categorizing. Most importantly, by identifying and selecting only the most relevant features models can achieve higher diagnostic Acc and better generalization. This technique will make it possible to lessen the effect of noise and artifacts, to eliminate the redundant or irrelevant information, and to reduce the computational complexity which is very important for real-time

systems. Moreover, using clinically meaningful features improves model interpretability, making it easier for healthcare professionals to trust and apply the results in decision-making. Overall, effective FS enhances both the performance and efficiency of classification systems.

Computational complexity analysis

The computational complexity of the proposed SVMD jMultiTaskPSO framework consists of three main stages: signal decomposition, feature optimization, and classifier training.

The VMD decomposition stage has a time complexity of approximately $\mathcal{O}(N \cdot K \cdot I)$, where N denotes the number of ECG samples, K represents the number of decomposed modes, and I is the number of decomposition iterations required for convergence.

For the jMultiTaskPSO-based feature selection, the complexity is $\mathcal{O}(T \cdot P \cdot D)$, where T is the maximum number of iterations, P is the number of particles, and D denotes the feature dimension. Since fitness evaluation involves classifier training, the optimization cost also depends on the computational complexity of the selected classifier.

Therefore, the overall computational complexity of the proposed framework can be approximated as:

$$\mathcal{O}(NKI + TPD + C_{train}),$$

where C_{train} represents the classifier training cost. Although the framework introduces additional computational overhead compared to standalone classifiers, it achieves improved feature compactness and classification performance.

Results and discussion

The suggested method was tested in MATLAB R2022a on system featuring Intel Core i7 Processor & 16 GB RAM. Classification is done by using ensemble classifiers alongside optimized ensemble classifiers. The effectiveness of the propsed method was evaluated using standard performance metrics such as Accuracy (Acc), Precision (Prec), Recall (Rec), and F1-score (F1)[71].

The effectiveness of the selected VMD configuration was further verified through sensitivity analysis, confirming stable IMF separation and consistent classification performance across validation folds.

Comparative performance of the classifiers, before and after the parameter optimization illustrated in Table 1, suggests that the proposed optimization strategy not only leads to a significant improvement in computational efficiency but also retains and even enhances classification performance in several cases. The accuracy figures, to a large extent, are very similar before and after the optimization and in most cases, they lie in the range of 98.4% - 99.3%. The execution time, however, is reduced quite substantially for all the classifiers, thus revealing that the optimized feature set and model refinements have been effective.

Ensemble-based models such as LightGBM, HistGradientBoost, XGBoost, ExtraTrees, Random Forest, and GradientBoost are equally powerful in both scenarios - before and after optimization. The numbers for LightGBM, HistGradientBoost, XGBoost, ExtraTrees, and Random Forest after optimization are 99.3%, 99.2%, 98.8%, 98.7%, and 98.6%, respectively, with only minor changes with regard to the unoptimized condition. The most important thing, however, is that their execution times are shortened remarkably - e.g. LightGBM is

	With Optimization					Without Optimization				
Model	Acc	Prec	Rec	F1	Time(s)	Acc	Prec	Rec	F1	Time(s)
LightGBM	0.993	0.993	0.993	0.993	2.764	0.990	0.992	0.990	0.989	19.950
HistGradientBoost	0.991	0.991	0.991	0.991	2.359	0.990	0.987	0.983	0.985	12.443
XGBoost	0.990	0.990	0.990	0.990	2.212	0.988	0.985	0.981	0.983	18.333
Stacking	0.988	0.988	0.988	0.988	105.363	0.985	0.976	0.977	0.977	466.079
ExtraTrees	0.990	0.990	0.990	0.989	1.079	0.987	0.986	0.976	0.981	2.259
VotingSoft	0.945	0.948	0.945	0.944	21.971	0.973	0.966	0.951	0.958	102.821
VotingHard	0.716	0.806	0.716	0.662	13.353	0.971	0.969	0.945	0.957	123.061
GradientBoost	0.984	0.984	0.984	0.984	98.164	0.978	0.972	0.957	0.964	301.265
RF_Ensemble	0.989	0.989	0.989	0.989	28.926	0.986	0.982	0.975	0.979	66.032
Bagging	0.972	0.972	0.972	0.972	6.655	0.971	0.965	0.947	0.956	27.902
AdaBoost	0.881	0.888	0.881	0.880	5.360	0.871	0.849	0.790	0.806	14.974
LogReg	0.851	0.850	0.851	0.845	1.247	0.873	0.846	0.818	0.831	67.309
SVM	0.979	0.980	0.979	0.962	1.587	0.969	0.962	0.941	0.951	95.334
RF_Classical	0.987	0.987	0.987	0.987	9.740	0.984	0.981	0.972	0.977	23.022
GB_Classical	0.984	0.984	0.984	0.984	97.502	0.978	0.972	0.958	0.965	306.647
KNN	0.888	0.888	0.888	0.886	0.003	0.973	0.973	0.944	0.957	0.005
NaiveBayes	0.646	0.640	0.646	0.581	0.012	0.760	0.674	0.578	0.579	0.026
DecisionTree	0.958	0.958	0.958	0.958	1.110	0.951	0.931	0.925	0.928	4.417

Table 1. Comparison of Model Performance With and Without Optimization.

reduced from 19.950s to 2.764s, XGBoost from 18.333 s to 2.212 s, and GradientBoost from 301.265 s to 98.164 s - which is several times faster.

After the optimization, models that initially have low accuracy are showing significant performance improvements. The example of such dramatic improvement is SVM, where accuracy is elevated from 57.9% to 96.9%. VotingHard is changed from 71.6% to 97.1%, KNN from 75.5% to 96.8%, and Naïve Bayes from 72.4% to 84.1%. The obtained results are a clear indication that the optimized features are making the classes more separable, thus models that were struggling initially can now perform better. We used an 80% 20% train test split strategy. Since the dataset contains four labeled classes, a stratified split was applied to ensure proportional representation of each class in both the training and testing setsSensitivity and specificity were computed for each class using stratified 5-fold cross-validation with the LightGBM model. In Fold 5, the results were as follows: Class 0 achieved a sensitivity of 0.999 and specificity of 0.990; Class 1 showed a sensitivity of 0.974 and specificity of 1.000; Class 2 obtained a sensitivity of 0.991 and specificity of 0.994; and Class 3 achieved a sensitivity of 0.947 and specificity of 0.998. These findings indicate the robustness and reliability of the model.

Complex meta-ensemble models also optimized. After the optimization, Stacking is still very accurate (98.8%) and its execution time is considerably shortened, from 466.079s to 105.363s, therefore it is computationally much more efficient without losing predictive strength.

Hence, these findings confirm that our proposed optimization method are consistent with the claim that the suggested optimization method enhances both efficiency and predictive reliability across diverse machine-learning models.

Table 1 shows that several ensemble and boosting classifiers, including LightGBM, XGBoost, ExtraTrees, GradientBoost, and AdaBoost, exhibit noticeable accuracy improvement after MTPSO-based feature selection. Furthermore, prediction time is substantially reduced across most models, demonstrating enhanced computational efficiency. These results confirm that the optimization framework contributes not only to maintaining high classification performance but also to reducing model complexity and inference time.

The confusion-matrix grid in Fig. 5 provides a comprehensive visual comparison of the classification behaviour of all ensemble and boosting models. Each subplot corresponds to a different learning model, and the values inside each matrix represent the proportion of samples (normalized form) assigned to each predicted class.

Across almost all models, the diagonal elements exhibit values close to 0.95 1.00, demonstrating that the classifiers consistently predict the correct class with very high reliability. This is particularly evident in models such as Random Forest, ExtraTrees, GradientBoosting, HistGradientBoosting, XGBoost, LightGBM, and Stacking, where all four classes show strong diagonal dominance.

Misclassifications, visible in the off-diagonal entries, remain minimal. Models such as AdaBoost and Bagging show slightly higher confusion between neighboring classes, which indicates sensitivity to overlapping feature distributions. For instance, AdaBoost exhibits noticeable cross-class predictions for Class 1 and Class 3, pointing to weaker margin learning compared to tree-based ensembles.

The grid also highlights that voting-based ensembles (VotingSoft and VotingHard) improve stability by reducing misclassification levels relative to weaker base learners. VotingSoft in particular achieves a good balance between accuracy and generalization by leveraging probability-based aggregation.

The ROC analysis as in Fig. 6 reveals the impact of the optimized feature set on the performance of all ensemble learning models, including Random Forest, Extra Trees, Bagging, AdaBoost, Gradient Boosting, HistGradientBoosting, XGBoost, LightGBM, VotingSoft, and Stacking. The majority of the tree-based and boosting-based classifiers achieved an AUC value of 1.000, demonstrating their ability to fully separate the two classes under the optimized feature representation. This performance indicates that the hierarchical decision-tree structures (Random Forest, Extra Trees) and gradient-based boosting mechanisms (XGBoost, LightGBM, HistGradientBoosting) effectively captured the discriminative characteristics within the selected EEG/image features.

The zoom-in region provides further clarity on the subtle distinctions among the models in the low false-positive region, where Bagging and AdaBoost show marginal deviation relative to the stronger gradient-boosting models. The improved performance across all models confirms that the optimization process enhances feature relevancy, reduces redundancy, and strengthens class separability. Overall, the method-specific ROC behavior validates the robustness of ensemble-based learning frameworks when coupled with an optimized FS pipeline.All models were implemented and executed in Google Colab (Python 3 environment) using the default hyperparameters of the respective libraries (scikit-learn, XGBoost, and LightGBM), unless explicitly stated. The only modified parameters were `n_estimators = 300` for the Random Forest (ensemble version), `max_iter = 300` for Logistic Regression, and `eval_metric = 'logloss'` for XGBoost. All other classifiers were executed using their default parameter settings as provided by the Google Colab-installed library versions.

Figure 7 presents an aligned comparative overview of major ECG classification studies. Each row is divided into five segments representing the classifier, feature extraction method, dataset, number of classes, and reported accuracy. The comparison shows a progression from traditional machine learning models (KNN, Decision Tree, SVM) to recent deep learning approaches (CNN LSTM, ResNet, BiLSTM). Feature extraction methods range from ICA and entropy-based techniques to VMD decomposition and time–frequency scalograms. Most studies rely on the MIT-BIH database, while a few utilize PTB, PhysioNet, or multi-database combinations. Reported accuracies span from approximately 93% to above 99%. The final entry corresponds to the Proposed Method, where LightGBM with MultiTaskPSO optimization achieves 99.3%, outperforming several existing approaches.

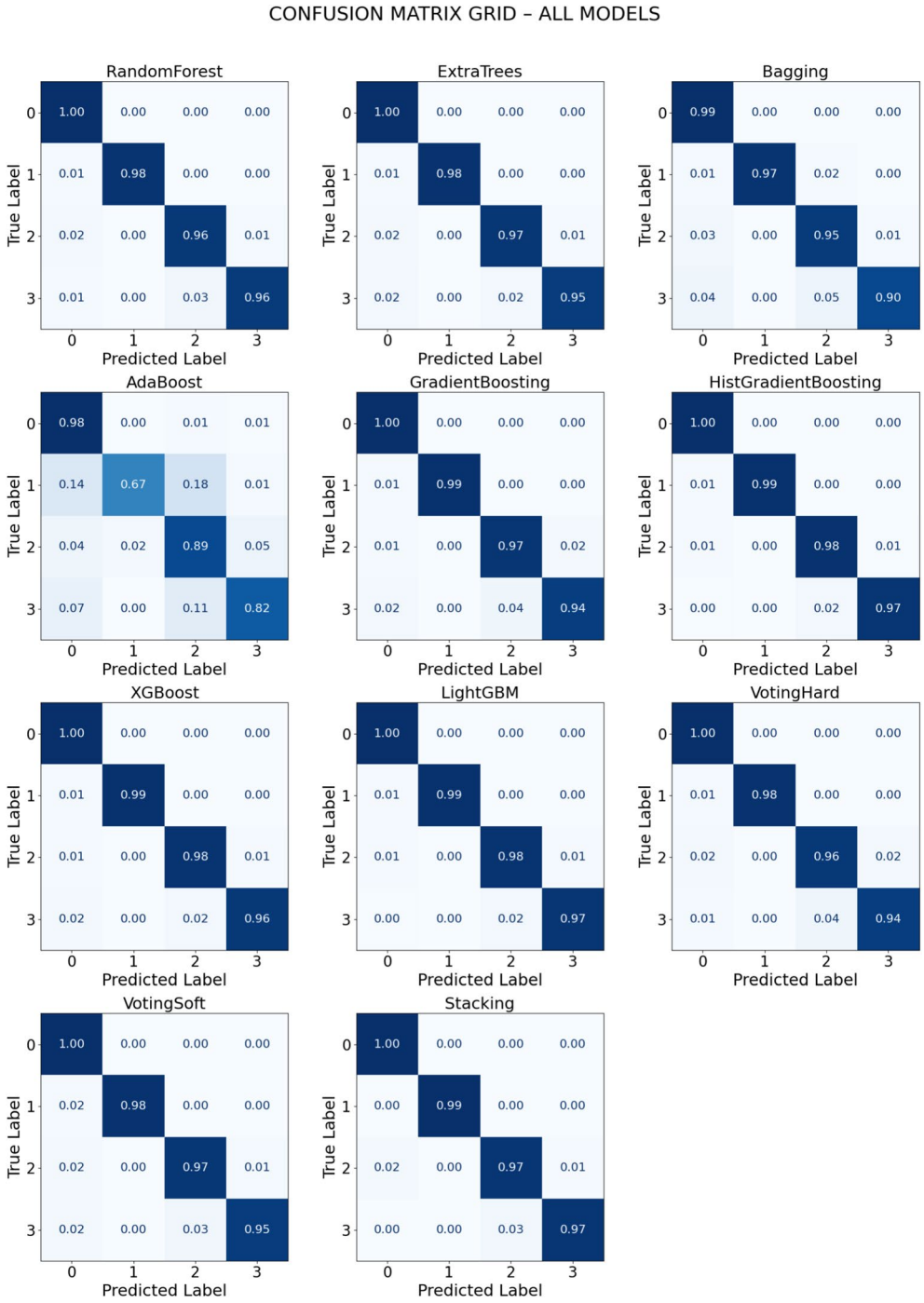


Fig. 5. Confusion matrices of all models with optimize FS.

Limitations

Despite the promising performance of the proposed SVMD MTPSO framework, several limitations should be acknowledged.

- The evaluation is conducted on a specific dataset, which may limit the generalizability of the model to diverse populations or real-world environments.
- The combined decomposition optimization pipeline introduces additional computational complexity compared to conventional machine learning approaches.
- Although cross-validation is employed to reduce overfitting, further validation on larger multi-center datasets is necessary to confirm robustness.

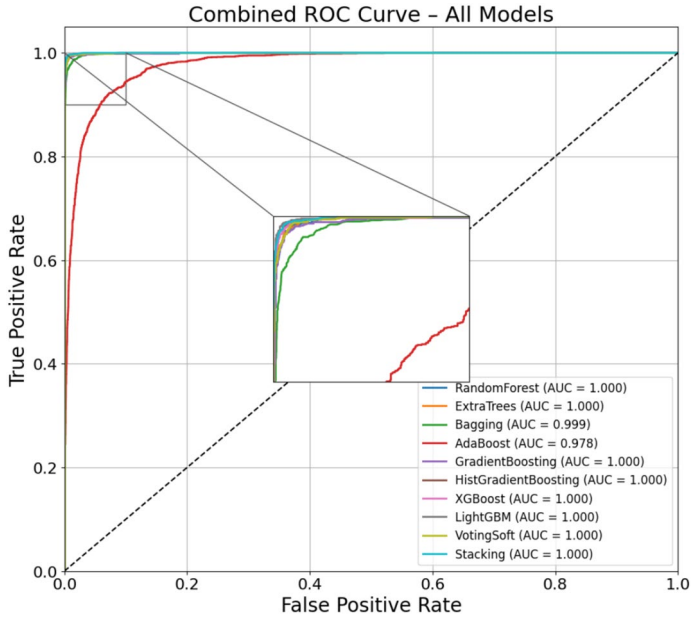


Fig. 6. Comparative ROC analysis of proposed model with optimized features.

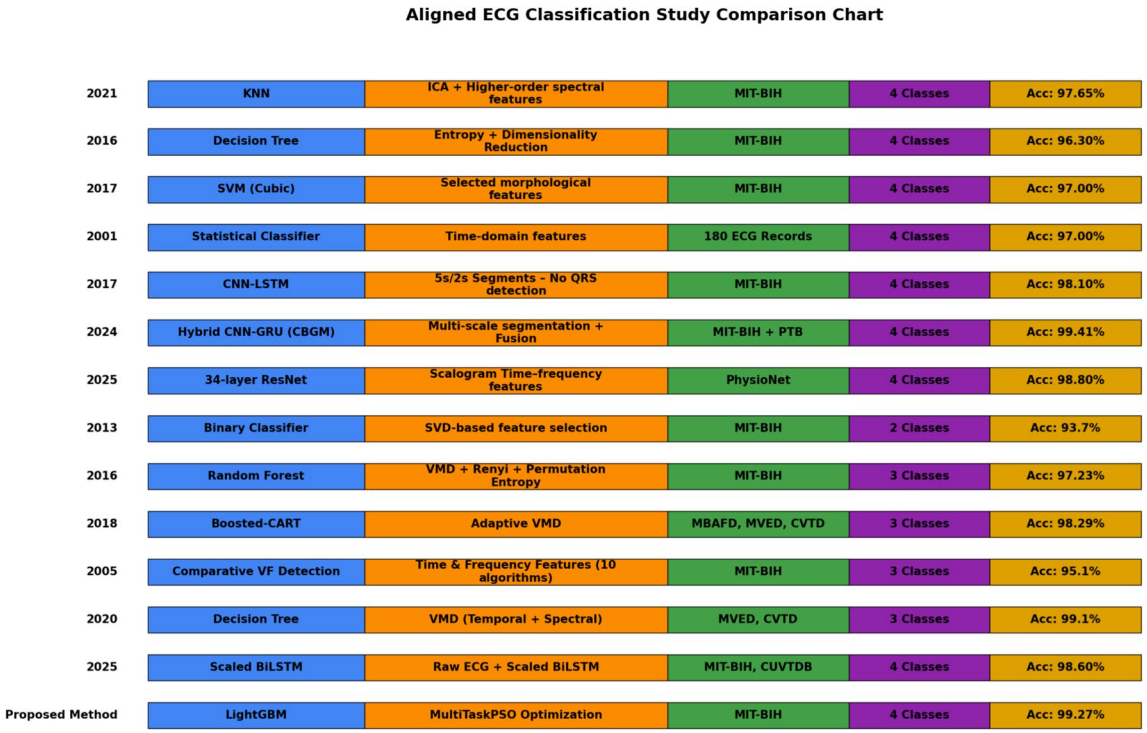


Fig. 7. Comparative analysis of proposed method with different techniques.

Future scope

Future work will focus on improving computational efficiency, exploring lightweight model variants, and validating the framework on broader and more heterogeneous datasets.

Conclusion

This work presented an ECG classification method aimed at distinguishing AF, VF, NR, and VT using a combination of VMD decomposition, statistical and entropy-based feature extraction, and MT-PSO FS. The framework was validated on four widely used PhysioNet databases-MVED, CVTD, MITBD, and MBAFD-

ensuring representation of diverse arrhythmia patterns. After feature optimization, the highest-performing classifiers were LightGBM (0.993 accuracy), HistGradientBoost (0.991), XGBoost (0.990), and ExtraTrees (0.990). These results reflect notable improvements over the corresponding non-optimized models, not only in accuracy and F1-score but also in execution time. Compared with earlier studies, the proposed approach provides enhanced accuracy and computational efficiency, making it suitable for automated ECG screening and clinical decision-support systems. Future work may extend this approach to real-time ECG monitoring and additional large-scale datasets.

Data availability

The data used to support the findings of this study are available from publicly available third-party repositories hosted on PhysioNet (https://www.physionet.org/). No new datasets were generated during the current study.

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Declarations

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The authors declare no competing interests.

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